# Chenyuan Zhou

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## Education

**Shanghai Jiao Tong University** 2023.9 - 2027.6(expected)
B.S in computer science, **ACM Honors Class**

GPA(Core Courses): **3.96**/4.3
**Selected courses**:

- Algorithm Design and Analysis :**A+, 98/100**
- Compiler Design and Implementation:**A+, 99/100**
- Computer Architecture:**A+, 97/100**
- Data Structure:**A+, 97/100**
- Mathematical Analysis(Honor):**A+, 95/100**

## Research interest

My research interests are Robotics, Multimodal Learning, and Test-Time Learning. I am especially interested in building multimodal systems that can efficiently understand spatial information, follow instructions, complete embodied tasks, and continually learn from new experiences in the real world.

## Research Experience

**Stanford CS, Stanford University (Yejin Choi, Leonidas Guibas)** 2026.6 - now

Research intern conducting experiments and participating in research on test-time adaptation of embodied video generation models, co-advised by Yejin Choi and Leonidas Guibas.

**RHOS (Yong-Lu Li, Ce-Wu Lu)** 2025.6 - 2026.6

Worked on precise manipulation with visual instruction and diverse tool adaptation under the supervision of Yong-Lu Li and Ce-Wu Lu.

## Projects

**Partial Rollout for LLM RL** (LLM Course Project) Spring 2025 Link to Repo

Decomposed the single rollout process into multiple turns and propagated unfinished rollouts to the next iteration, reducing the negative impact of long-tail rollouts.
- Achieved a 30% increase in overall speed without performance degradation.
- Implemented partial code and conducted experiments in collaboration with Zeng Ji and Li Zhiyan. Details in PR 1826

**RISCV CPU** (Course Project) Fall 2024 Link to Repo

Designed a CPU in Verilog implementing the basic RISCV instruction set.
- Implemented branch prediction and instruction prefetching.
- Successfully executed on FPGA.

**Mx Compiler** (Course Project) Summer 2024 Link to Repo

A compiler for Mx* (an educational language with basic features of C), as well as Clang (with `mem2reg` and register allocation).
- Implemented various optimizations including SCCP, DCE, inlining, GVN & GCM, loop detection, and unrolling.
- Utilized SSA graph coloring to allocate registers based on liveness analysis.

# Papers

### T2-z: Test-Time Optimization of Environment Latents for Robotic World Models

**Chenyuan Zhou**, Yining Hong, Huang Huang, Yunfan Jiang, Wenlong Huang, David Held, Vitor Campagnolo Guizilini, Yue Wang, Li Fei-Fei, Jiajun Wu, Leonidas Guibas, and Yejin Choi.

*arXiv:2610.22364, 2026* Paper | Project | Code

Introduced T2-z, an adaptive robotic world model that captures unobserved environmental factors in a learned environment latent.

- Adapted to unseen environments by optimizing only the latent from test-time interactions while keeping the world model frozen.

- Evaluated across seven simulated settings and four real-world robotic tasks, reducing object prediction error from 15.20 to 7.00 and improving action success from 33.73% to 72.96%.

### $A^2$-VLA: Adapting Vision-Language-Action Policies to New Tools through Agentic Affordance Discovery

*(In submission to CVPR 2027)*

Developed $A^2$-VLA, a human-inspired tool affordance learning framework that captures transferable tool-use priors and enables generalization to unseen tools, as a co-leader.

- Used embodied agents to model tool affordances from unguided exploration, tool observation, and task-directed demonstrations.

- Used autonomously distilled affordances to train and guide vision-language-action (VLA) models, improving robotic manipulation performance and generalization to novel tools.

# Awards and Honors

**2023, 2024 Zhiyuan Honors Scholarship** (**2 %** in SJTU)

# Student Work and Teaching Experience

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|---|---|
| **Computer Programming, Teaching Assistant** | 2024 Fall |
| **Data Structure, Teaching Assistant** | 2025 Spring |
| **Principle and Practice of Computer Algorithms(AI), Teaching Assistant** | 2025 Summer |
| **Vice Monitor of ACM Honors Class 2024** | 2024.9 - 2025.9 |

# Skills

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|---|---|
| **Programming** | C++, Python, Java, Verilog, |
| **Tools** | Git, Solidworkers, ros2, LaTeX |
| **Languages** | Chinese (Native speaker), English (TOEFL iBT: 5.5/6) |